



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Nanyang Business School

ROBOTIC PROCESS AUTOMATION

Working with Data

Data tables, spreadsheets and regular expressions

01

DataTable Structure

Rows, columns, and the copy held in memory.



02

Reading Data

Read Range, the header setting, and reading a value from a row.

03

Writing Data

Writing back to a sheet that already holds data.

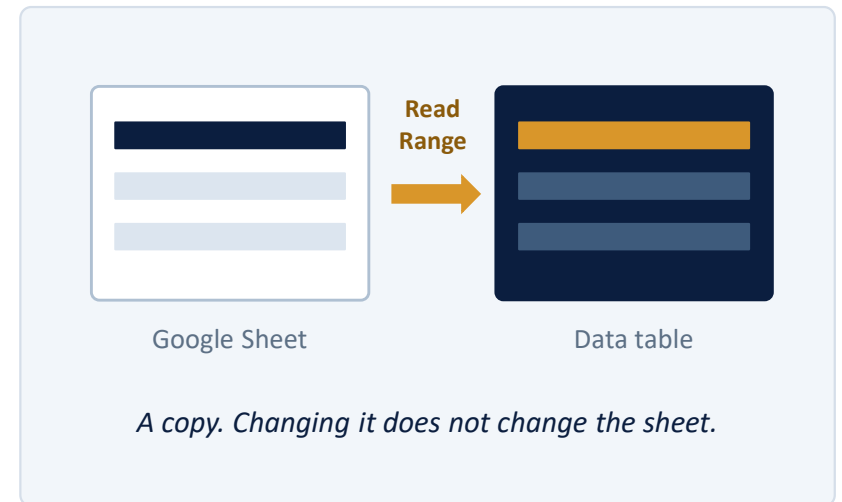
04

Regular Expressions

Validating, extracting and cleaning text with patterns.

A data table is a copy held in memory

- When the robot reads a spreadsheet range, it does not stay connected to the sheet. It takes a copy.
- That copy is a data table. It has rows and columns like the sheet, but it is held in the robot's memory.
- Nothing you change in the copy appears in the sheet until you write it back.



PITFALL

Junior developers often expect a data table to behave like a live link to the sheet. It does not. If your automation finishes and the sheet looks unchanged, the usual cause is that you edited the copy and never wrote it back.

The parts of a data table

ROW INDEX | COLUMN NAMES

	FirstName	Email	Status
0	Aisha	aisha.k@corp.com	
1	Ben	ben@@corp.com	
2		chan.l@corp.com	

PITFALL

The first row of data is index 0, not 1. The header row is not counted at all. This is exactly where the plus two in your exercise comes from, and we come back to it in section three.

- Columns have a name and a data type. The default type is text.
- Rows are numbered from zero.
- The header row is not a row in the it is the column names.

01

DataTable Structure

Rows, columns, and the copy held in memory.

02

Reading Data

Read Range, the header setting, and reading a value from a row.



03

Writing Data

Writing back to a sheet that already holds data.

04

Regular Expressions

Validating, extracting and cleaning text with patterns.

Read Range

users ☆ 📁 ☁

File Edit View Insert Format Data

🔍 ↶ ↷ 🖨 📄 100% ▼ £ %

I15 ▾ *fx*

	A	B	C
1	FirstName	Email	Status
2	Aisha	aisha.k@corp.com	
3	Ben	ben@corp.com	
4	Chan	chan.l@corp.com	
5			
6			

Main.xaml

Manual Trigger

Output

Job data

Output

read range

- Read Range** ⌚
Reads and saves the information from a spreadsheet's range for subseque... Google Workspace
- Read Range**
Reads and saves the information from an Excel Workbook's range for subsequen... Microsoft 365
- Read Range**
Reads data from a specified range in an Apple Numbers spreadsheet Apple Numbers
- Read Range Workbook**
Reads the value of a range in Range Excel Format from a spreadsheet as a DataTable. If Ra... Excel

Read Range

Main.xaml

⚡ TRIGGER

Manual Trigger

📄 Read Range

Read ...

Google Sheets connection

Spreadsheet *
users

Range *
Sheet1 (Sheet)

Has headers
☒ True ☐ False

Input

What to read
Values

Output

Read range Output
Read range

Range information Output
Range information

The range information as DataTable Automatically generated output variable. You can use it as it is or save it with a different name.

Use

{x} Change variable

✕ Create variable

Ⓐ Create argument

Reading a value from a row

Main.xml

⚡ TRIGGER

👤 Manual Trigger

📄 Read Range

📄 For Each Row in Data Table

For Each Row in Data ...

Data Table*

Click to use a variable

🔍 Search

▼ 📄 Read Range

- 📄 Read range
- 📄 Range information
- ☰ Columns
- ☰ Rows
- > [Show more](#)

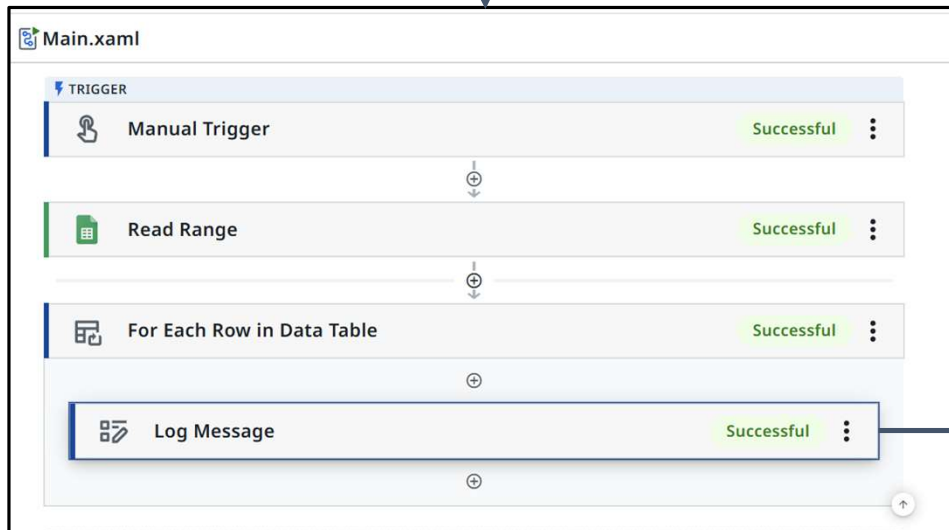
> 👤 Manual Trigger

Index Output

📄 Index

Reading a value from a row

	A	B	C
1	FirstName	Email	Status
2	Aisha	aisha.k@corp.com	
3	Ben	ben@corp.com	
4	Chan	chan.l@corp.com	



VB Expression editor Log Message > Message ({} Object)

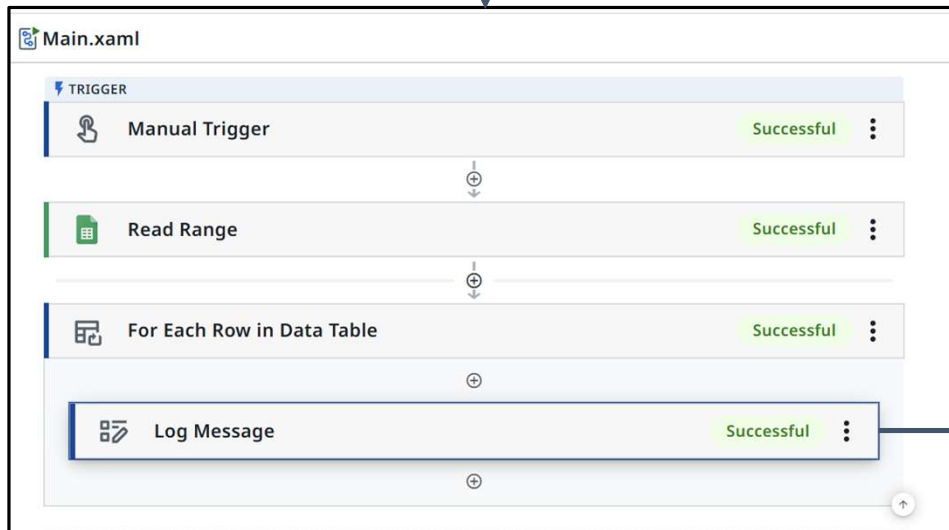
1 CurrentRow("FirstName").ToString

► Debug on cloud

- 12:45:04 PM Building project completed
- 12:45:05 PM RPA Workflow execution started
- 12:45:07 PM Aisha
- 12:45:07 PM Ben
- 12:45:07 PM Chan
- 12:45:07 PM RPA Workflow execution ended

Reading a value from a row

	A	B	C
1	FirstName	Email	Status
2	Aisha	aisha.k@corp.com	
3	Ben	ben@corp.com	
4	Chan	chan.l@corp.com	



VB Expression editor Log Message > Message ({ } Object)

Log Message > Message ({ } Object)

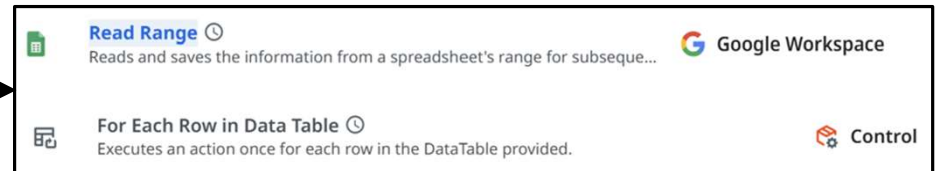
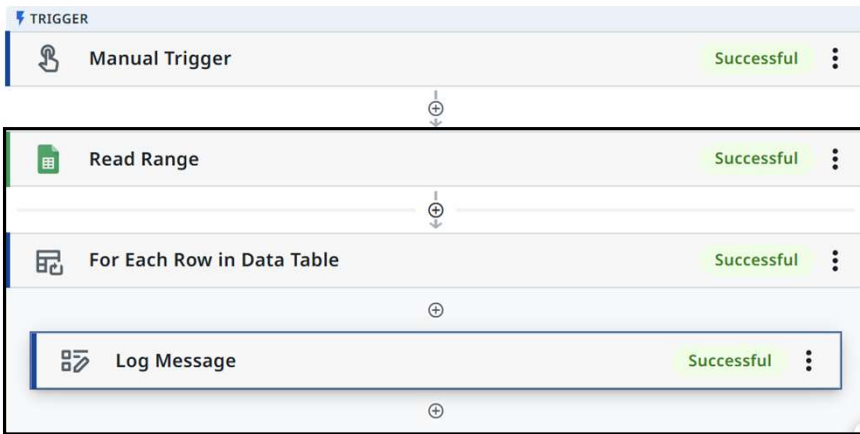
1 CurrentRow(0).ToString

► Debug on cloud

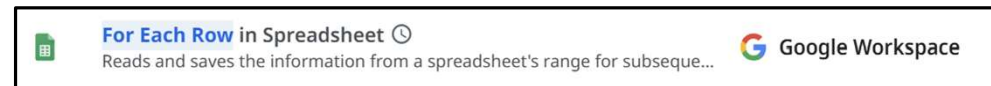
- 12:45:04 PM Building project completed
- 12:45:05 PM RPA Workflow execution started
- 12:45:07 PM Aisha
- 12:45:07 PM Ben
- 12:45:07 PM Chan
- 12:45:07 PM RPA Workflow execution ended

Reading a value from a row

Main.xml



=



01

DataTable Structure

Rows, columns, and the copy held in memory.

02

Reading Data

Read Range, the header setting, and reading a value from a row.

03

Writing Data

Writing back to a sheet that already holds data.



04

Regular Expressions

Validating, extracting and cleaning text with patterns.

Writing to existing table

Before

	A	B	C
1	FirstName	Email	Status
2	Aisha	aisha.k@corp.com	
3	Ben	ben@corp.com	
4	Chan	chan.l@corp.com	

After

	A	B	C
1	FirstName	Email	Status
2	Aisha	aisha.k@corp.com	OK
3	Ben	ben@corp.com	OK
4	Chan	chan.l@corp.com	OK

Writing to existing table

For Each Row in Data Table

Successful

Data Table*

Read range

×

Item name*

CurrentRow

Show additional properties

▼

⊕

Assign

Successful

To variable*

Output

{*} CurrentRow("Status")

×

Set value *

OK

@

⊕

↑

assign

×

Assign

ⓘ

Assign a value to a variable, enabling its reuse in subsequent activities within the workfl...

Control

Writing to existing table

For Each Row in Data Table

Data Table*

Read range X

Item name*

CurrentRow

Show additional properties

Assign

To variable* Output

{x} CurrentRow("Status") X

Set value *

OK

1. The modified datatable "Read range" is being written

Write Range Successful

Google Sheets connection

tanyongheng@gmail.com X

Spreadsheet *

users

Range *

Sheet1 (Sheet)

Include headers

☒ True ☐ False

How to write

Overwrite

What to write*

Read range X

2. Besides "overwrite", there are also "append" and "insert" options

- Overwrite**
Overwrite any previous data that exists
- Append**
Append to the Bottom
- Append Right**
Append to the Right
- Insert**
Insert a Row

01

Data Table Structure

Rows, columns, and the copy held in memory.

02

Reading Data

Read Range, the header setting, and reading a value from a row.

03

Writing Data

Writing back to a sheet that already holds data.

04

Regular Expressions

Validating, extracting and cleaning text with patterns.



If we want to check a variable against a single value

The diagram illustrates the process of configuring a conditional logic block. It starts with an 'If' block containing a 'Then' section with the action 'do something' and an 'Else' section. A 'Condition *' block is shown next to it, containing a placeholder '{x} EmailAddress...'. An arrow points from this placeholder to a 'Condition builder' panel. The 'Condition builder' panel allows selecting logical operators ('All (AND)' or 'Any (OR)') and defining a condition where '{x} Email address' is 'equals' to the value 'michael.tan@ntu.edu.sg'.

If

- ✓ Then
 - // do something
- ✗ Else

Condition *

{x} EmailAddress...

Condition builder

✓ All (AND) Any (OR) of the conditions are met:

{x} Email address × equals michael.tan@ntu.edu.sg

Add ▾

VALIDATING FORMAT

What if we want to check against a certain format?

Not just "is there an @ sign" — but does the whole string follow a defined shape?

`firstname.lastname`

Two words, separated by a full stop

@

Exactly one at sign

`domain.com`

A domain name and an extension

Matches the format

- `jane.tan@company.com`
- `wei.lim@acme.com`
- `mary.ng@school.edu`

Does not match

- `jane@company.com`
- `jane.tan@company`
- `jane.tan.company.com`

Using Regular Expression (RegEx) to identify email address format



System.Text.RegularExpressions.IsMatch(Email, "\w + \. + \w + @ \w + \. \w + \$") equals true

VB Expression editor If > Condition (Boolean)

Fix {x} Insert variable

```
1 System.Text.RegularExpressions.Regex.IsMatch(CurrentRow.Email, "\w + \. + \w + @ \w + \. \w + $")
```

An email pattern

```
^[\\w\\.]+@[\\w+\\.\\w+]+$
```

^

start of the text

[\\w\\.]+

one or more letters, digits, underscores or dots

@

a literal at sign

\\w+

one or more letters or digits

\\.

a literal dot

\\w+

one or more letters or digits

\$

end of the text

NOTE

Read the pattern from left to right. Start of text, name characters, an at sign, domain characters, a dot, further characters, end of text.

Email addresses that pass and fail

WHAT THE CHECK DECIDED

aisha.k@corp.com Passed

ben@corp.com Passed

finance@sub.corp.co.uk Failed

oneill-smith@corp.com Failed

THE PATTERN IT USED

```
"^[\\w\\.]+@\\w+\\.\\w+$"
```

Can you identify the cause of the failures?

Common regex symbols

Character types

`\d` any digit

`\w` letter, digit or underscore

`\s` any whitespace

`.` any character at all

How many

`+` one or more

`*` zero or more

`?` zero or one, so optional

`{2,4}` between two and four

Sets and position

`[abc]` any one of a, b or c

`[^abc]` any character except those

`[a-z]` any character in the range

`^` `$` start and end of the text

PITFALL

A dot on its own means any character, so a pattern written as `\w+.\w+` will accept an email with no dot in it. To mean an actual full stop, escape it as backslash dot. The same applies to the other symbols that have a job in a pattern, including plus, star, question mark, brackets and the dollar sign.

Useful patterns

What you want	Pattern
Digits only, nothing else	<code>^\d+\$</code>
An amount, with optional cents	<code>^\d+(\.\d{2})?\$</code>
A date written as 2026-04-30	<code>^\d{4}-\d{2}-\d{2}\$</code>
A reference like INV-00123	<code>^INV-\d{5}\$</code>
A Singapore mobile or landline	<code>^[689]\d{7}\$</code>
Any text at all, but not blank	<code>^\s*\S.*\$</code>

TIP

Test a pattern before building it into a workflow. A free site such as [regex101](https://regex101.com/) shows which characters matched and why. This is faster than testing by trial and error in Studio.

Key Lessons

01

DataTable Structure

A data table is a copy in memory, remember to write it back

02

Reading Data

Use column names or index to access the cell

03

Writing Data

Check the header setting

04

Regular Expressions

Rigorous testing needed